Lecture 25: Gaussian Quadrature

Math 6610

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Today's Goals

- Free nodes: the best 2-point rule
- Any interval $[a, b]$
- Orthogonal (Legendre) polynomials give the nodes for n points
- G_n is exact on $\prod_{2n-1}$, and this is optimal
- Nodes and weights in practice

Degree of Exactness

A quadrature rule adds weighted values of f:

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i). \tag{1}$$

Definition (*Degree of exactness*). Rule (1) has *degree of exactness* m if it is exact for all polynomials of degree $\leqslant m$ (the set $\prod_m$), but not for all of degree $m+1$.

Next: how are the nodes and weights chosen?

The Recipe

How the weights are found (Lecture 23).

1. Fix n nodes $x_1, \ldots, x_n$.
2. Replace f by its interpolating polynomial $p = \sum_i f(x_i)\,\ell_i$, $\ell_i = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$.
3. Integrate p: $\int_a^b f\,dx \approx \int_a^b p\,dx = \sum_i f(x_i) \int_a^b \ell_i\,dx$, so $w_i = \int_a^b \ell_i\,dx$.

If $f \in \prod_{n-1}$ then $p = f$: degree of exactness $\geqslant n - 1$.

Trapezoid

Check the recipe on a rule you know: which weights, which degree of exactness? Interval $[-1, 1]$, nodes chosen first.

Nodes $-1, 1$ ($n = 2$): $\ell_1 = \frac{1-x}{2}$, $\ell_2 = \frac{1+x}{2}$,

$$w_1 = \int_{-1}^{1} \frac{1-x}{2}\, dx = 1, \quad w_2 = 1, \qquad R(f) = f(-1) + f(1).$$

Exact on $\prod_1$; but x^2 gives $1 + 1 = 2 \neq \frac{2}{3}$: degree 1.

Simpson

Same recipe, $[-1, 1]$.

Nodes $-1, 0, 1$ ($n = 3$): $\ell_1 = \frac{x(x-1)}{2}$, $\ell_2 = 1 - x^2$, $\ell_3 = \frac{x(x+1)}{2}$,

$$w_1 = w_3 = \tfrac{1}{3}, \quad w_2 = \int_{-1}^{1} (1 - x^2)\, dx = \tfrac{4}{3}, \qquad R(f) = \tfrac{1}{3}\big[f(-1) + 4f(0) + f(1)\big].$$

Exact on $\prod_2$, and on x^3 too ($-\frac{1}{3} + \frac{1}{3} = 0$ by symmetry); x^4 gives $\frac{2}{3} \neq \frac{2}{5}$: degree 3.

Pattern: the nodes were our choice; only the weights were computed.

Free Nodes

In the recipe the nodes came first; only the weights were computed.

Idea. Make the nodes unknowns too. Two points on $[-1, 1]$:

$$\int_{-1}^{1} f(x)\,dx \approx w_1 f(x_1) + w_2 f(x_2), \qquad \text{unknowns } w_1, w_2, x_1, x_2.$$

- 4 unknowns; a cubic has 4 coefficients.
- Demand exactness on $1, x, x^2, x^3$: 4 equations.
- Trapezoid: degree 1 with 2 values. Can 2 values reach degree 3?

Two Points (1/2)

Proposition. The rule $R(f) := w_1 f(x_1) + w_2 f(x_2)$ is exact on $\prod_3$ if and only if

$$w_1 + w_2 = 2, \qquad w_1 x_1 + w_2 x_2 = 0,$$
$$w_1 x_1^2 + w_2 x_2^2 = \tfrac{2}{3}, \qquad w_1 x_1^3 + w_2 x_2^3 = 0.$$

Proof. 1. Reduce to monomials. For $p = c_0 + c_1 x + c_2 x^2 + c_3 x^3$,

$$R(p) - \int_{-1}^{1} p\,dx = \sum_{k=0}^{3} c_k \left(w_1 x_1^k + w_2 x_2^k - \int_{-1}^{1} x^k\,dx \right).$$

This vanishes for all $c_0, \ldots, c_3$ iff each bracket is 0 (take $c = e_k$).
2. The integrals. $\int_{-1}^{1} x^k dx = 2, 0, \frac{2}{3}, 0$ for $k = 0, 1, 2, 3$. ■

Two Points (2/2)

Proposition. With $x_1 < x_2$, the only solution is $x_1 = -\frac{1}{\sqrt{3}}$, $x_2 = \frac{1}{\sqrt{3}}$, $w_1 = w_2 = 1$.

Proof. 1. Weights are nonzero. If $w_2 = 0$, the first two equations give $w_1 = 2$, $x_1 = 0$, and the third reads $0 = \frac{2}{3}$. Same for w_1.
2. Use the odd equations. By the second equation $w_1 x_1 = -w_2 x_2$, so the fourth equation becomes

$$0 = w_1 x_1 \cdot x_1^2 + w_2 x_2^3 = w_2 x_2 \, (x_2^2 - x_1^2).$$

3. Symmetry. $w_2 \neq 0$. If $x_2 = 0$, then $w_1 x_1 = 0$, so $x_1 = 0 = x_2$: excluded. Hence $x_2^2 = x_1^2$, and $x_1 < x_2$ gives $x_1 = -s$, $x_2 = s$ with $s > 0$.
4. Values. Second equation: $s(w_2 - w_1) = 0$, so $w_1 = w_2$; first: $w_1 = w_2 = 1$; third: $2s^2 = \frac{2}{3}$, so $s = \frac{1}{\sqrt{3}}$. The fourth holds: $-s^3 + s^3 = 0$. ■

The Two-Point Gauss Rule

Definition (*Two-point Gauss rule on* $[-1,1]$*).*

$$G_2(f) := f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right), \qquad \frac{1}{\sqrt{3}} \approx 0.577350.$$

By the two propositions G_2 is exact on $\prod_3$. It is not exact on $\prod_4$: $G_2(x^4) = \frac{2}{9} \neq \frac{2}{5} = \int_{-1}^{1} x^4 dx$. So the degree of exactness is exactly 3.

Same cost as the trapezoid rule, degree 3 instead of 1.

Other Intervals (1/2)

Tables are for $[-1, 1]$. For $[a, b]$ put

$$\varphi(t) := \frac{b+a}{2} + \frac{b-a}{2}\, t, \qquad \varphi(-1) = a, \quad \varphi(1) = b, \quad \varphi' = \tfrac{b-a}{2} > 0.$$

Substituting $x = \varphi(t)$, $dx = \frac{b-a}{2}\, dt$:

$$\int_a^b f(x)\, dx = \frac{b-a}{2} \int_{-1}^{1} f(\varphi(t))\, dt. \tag{2}$$

Other Intervals (2/2)

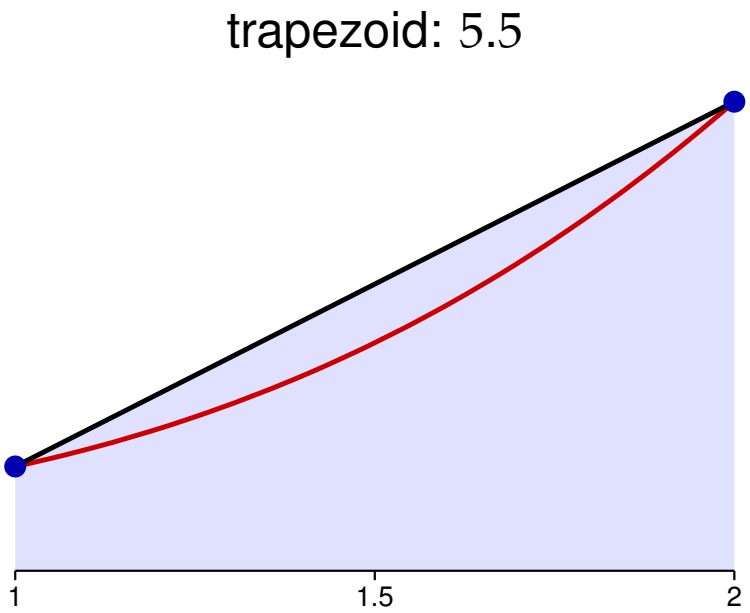
Proposition. Let $\sum_i \hat{w}_i\, q(t_i)$ be a rule on $[-1,1]$ with degree of exactness m. With $x_i := \varphi(t_i)$ and $w_i := \frac{b-a}{2}\hat{w}_i$, the rule $\sum_i w_i\, p(x_i)$ on $[a,b]$ has degree of exactness m.

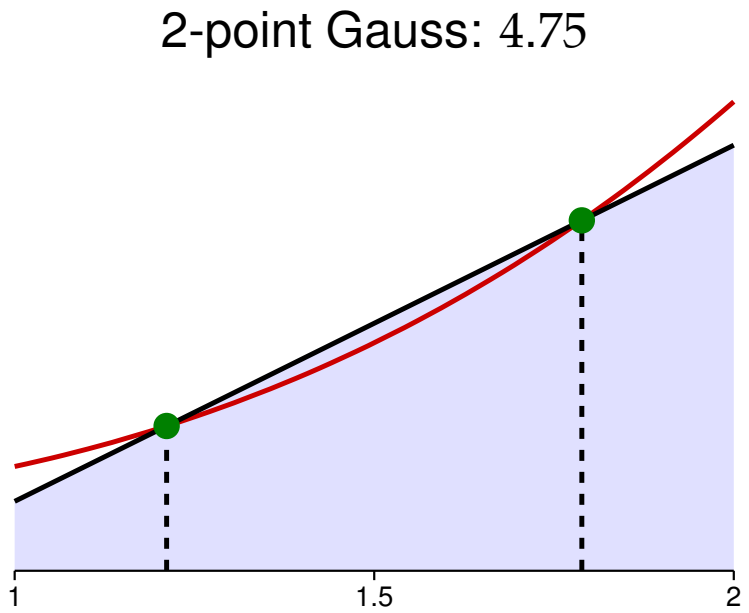
Proof. For $p \in \prod_m$, also $p \circ \varphi \in \prod_m$ (φ is affine), so by (2)

$$\int_a^b p\,dx = \frac{b-a}{2} \sum_i \hat{w}_i\, p(\varphi(t_i)) = \sum_i w_i\, p(x_i).$$

If it were exact on $\prod_{m+1}$ too, apply this to $q \circ \varphi^{-1}$ for $q \in \prod_{m+1}$: the rule on $[-1,1]$ would be exact on $\prod_{m+1}$, contradiction. ∎

Graphical View





$\int_1^2 (x^3+1)\,dx = \frac{15}{4} + 1 = 4.75$. G_2 on $[1,2]$: $x_{1,2} = \frac{3}{2} \mp \frac{1}{2\sqrt{3}} \approx 1.2113,\ 1.7887$, weights $\frac{1}{2}$:

$$G_2 = \tfrac{1}{2}\big[f(x_1) + f(x_2)\big] = \tfrac{1}{2}\big[2.7774 + 6.7226\big] = 4.75 = I.$$

G_2 is the area under the line through the two nodes (right panel).

Logarithm Example

Example. $\int_1^{1.5} x^2 \ln x \, dx$: $\varphi(t) = \frac{t+5}{4}$, $dx = \frac{1}{4}dt$, nodes $x_{1,2} = \frac{5}{4} \mp \frac{1}{4\sqrt{3}}$.

$$G_2 = \tfrac{1}{4}\left[x_1^2 \ln x_1 + x_2^2 \ln x_2\right] \approx 0.1922687.$$

Exact 0.1922594 (antiderivative $\frac{x^3}{3}\ln x - \frac{x^3}{9}$); error $9.3 \cdot 10^{-6}$.

G_2: two function values.

Exponential Example

Example. $\int_0^1 x^2 e^{-x}\,dx$: $\varphi(t) = \frac{t+1}{2}$, $dx = \frac{1}{2}dt$, nodes $x_{1,2} = \frac{1}{2} \mp \frac{1}{2\sqrt{3}}$.

$$G_2 = \tfrac{1}{2}\left[x_1^2 e^{-x_1} + x_2^2 e^{-x_2}\right] \approx 0.1594104.$$

Exact $2 - 5e^{-1} = 0.1606028$; error $-1.2 \cdot 10^{-3}$. Simpson (3 values): 0.162402, error $1.8 \cdot 10^{-3}$. Two Gauss values beat three Simpson values, although f is not a cubic.

More Points: The Key Idea

Let $p \in \prod_{2n-1}$. Divide by a degree-n polynomial q_n whose zeros will be the nodes:

$$p = q_n\, s + r, \qquad s, r \in \prod_{n-1}.$$

- At a node $p(x_i) = r(x_i)$: the rule sees only r.
- Weights exact on $\prod_{n-1}$ integrate r exactly.
- Need $\int q_n s\, dx = 0$ for all $s \in \prod_{n-1}$: q_n *orthogonal* to $\prod_{n-1}$.

Check: $n = 2$, $q_2 = 3x^2 - 1$ has zeros $\pm\frac{1}{\sqrt{3}}$ and

$$\int_{-1}^{1} (3x^2 - 1) \cdot 1\, dx = 0, \qquad \int_{-1}^{1} (3x^2 - 1)\, x\, dx = 0.$$

Orthogonal Polynomials

Definition (*Orthogonality; orthogonal polynomial*). For polynomials p, q let $\langle p, q\rangle := \int_{-1}^{1} p(x)q(x)\,dx$; p and q are *orthogonal* if $\langle p, q\rangle = 0$. The n*th orthogonal polynomial* is the monic (leading coefficient 1) $p_n(x) = x^n + \cdots \in \prod_n$ with

$$\langle q, p_n\rangle = 0 \qquad \text{for all } q \in \prod_{n-1}.$$

Construction: Gram–Schmidt on $1, x, x^2, \ldots,$

$$p_0 = 1, \qquad p_n = x^n - \sum_{j=0}^{n-1} \frac{\langle x^n, p_j\rangle}{\langle p_j, p_j\rangle}\, p_j.$$

The p_j, $j < n$, span $\prod_{n-1}$, so p_n is orthogonal to $\prod_{n-1}$. First three:

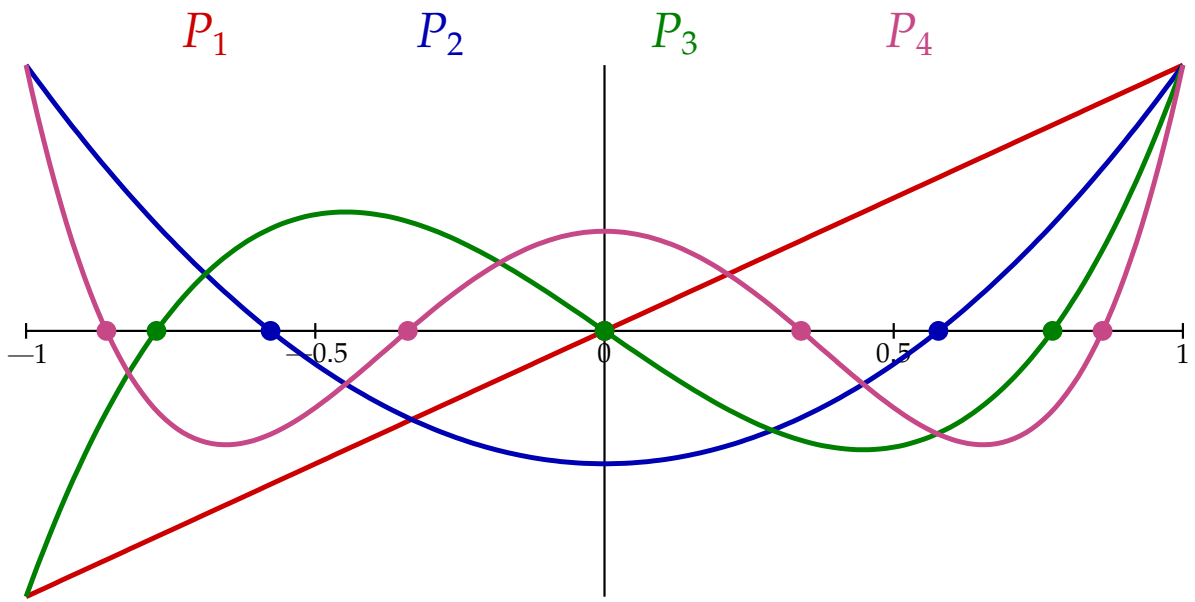
$$p_1 = x, \qquad p_2 = x^2 - \tfrac{1}{3}, \qquad p_3 = x^3 - \tfrac{3}{5}x.$$

(Check: $\int_{-1}^{1}(x^2 - \frac{1}{3})dx = 0$; $\int_{-1}^{1}(x^3 - \frac{3}{5}x)\,x\,dx = \frac{2}{5} - \frac{2}{5} = 0$.)

Legendre Polynomials

Definition (*Legendre polynomials*). $P_0 = 1$, $P_1 = x$, and for $n \geqslant 2$

$$P_n(x) = \frac{2n-1}{n}\, x\, P_{n-1}(x) - \frac{n-1}{n}\, P_{n-2}(x), \qquad P_2 = \frac{3x^2-1}{2}, \quad P_3 = \frac{5x^3-3x}{2}.$$



Standard fact (used without proof): the P_n are orthogonal to each other. $P_n(1) = 1$ and $\deg P_n = n$, so P_n is a constant multiple of p_n ($P_2 = \frac{3}{2}p_2$): *same zeros* (dots in the figure).

The Zeros of p_n

Lemma. For $n \geqslant 1$, p_n has n distinct real zeros, all in $(-1, 1)$.

Proof. 1. Sign changes. Let $z_1, \ldots, z_m$ be the points of $(-1, 1)$ where p_n changes sign. Then $m \leqslant n$, as $p_n \not\equiv 0$ has at most n zeros.
2. Assume $m \leqslant n - 1$. Put $q(x) := \prod_{i=1}^{m}(x - z_i) \in \prod_{n-1}$ ($q \equiv 1$ if $m = 0$).
3. One sign. Each factor $x - z_i$ changes sign exactly where p_n does, so the continuous $p_n q \not\equiv 0$ has one sign on $(-1, 1)$, and

$$\langle p_n, q \rangle = \int_{-1}^{1} p_n(x) q(x)\, dx \neq 0.$$

4. Contradiction. Orthogonality gives $\langle p_n, q \rangle = 0$. Hence $m = n$: n distinct zeros in $(-1, 1)$, and no others, since $\deg p_n = n$. ■

The Gauss Rule

Definition (*Gauss rule G_n*). Let $-1 < x_1 < \cdots < x_n < 1$ be the zeros of p_n, and

$$w_i := \int_{-1}^{1} \ell_i(x)\,dx, \qquad \ell_i(x) := \prod_{j \neq i} \frac{x - x_j}{x_i - x_j} \quad (i = 1, \ldots, n).$$

The *n-point Gauss rule* is $G_n(f) := \sum_{i=1}^{n} w_i f(x_i)$.

Remark (*Weights*). As in the recipe, G_n is exact on $\prod_{n-1}$. For $n = 2$: zeros of p_2 are $\pm\frac{1}{\sqrt{3}}$, $w_1 = w_2 = 1$: the G_2 found by hand.

Gauss Is Exact on $\prod_{2n-1}$

Theorem. $G_n(p) = \int_{-1}^{1} p(x)\,dx$ for all $p \in \prod_{2n-1}$.

Proof. 1. Divide. Let $p \in \prod_{2n-1}$. Polynomial division by p_n (degree n) gives s, r with

$$p = p_n s + r, \qquad s, r \in \prod_{n-1}.$$

2. Integrate. $\int_{-1}^{1} p_n s\,dx = \langle s, p_n \rangle = 0$, because $s \in \prod_{n-1}$. So $\int_{-1}^{1} p\,dx = \int_{-1}^{1} r\,dx$.

3. Exact on r. $r \in \prod_{n-1}$, so $\int_{-1}^{1} r\,dx = \sum_i w_i r(x_i)$.

4. Nodes are zeros. $p_n(x_i) = 0$, so $p(x_i) = r(x_i)$ for each i. Together:

$$\int_{-1}^{1} p(x)\,dx = \int_{-1}^{1} r(x)\,dx = \sum_{i=1}^{n} w_i r(x_i) = \sum_{i=1}^{n} w_i p(x_i).$$

■

Degree $2n-1$ Is Optimal

Proposition. No rule (1) with n nodes is exact on $\prod_{2n}$. Hence G_n has degree of exactness exactly $2n-1$.

Proof. Let $x_1, \ldots, x_n$ be the nodes of any such rule and $q(x) := \prod_{i=1}^{n}(x-x_i)^2 \in \prod_{2n}$. Then $q \geqslant 0$ is continuous and $q \not\equiv 0$, so $\int_{-1}^{1} q\,dx > 0$, while $\sum_i w_i q(x_i) = 0$. ∎

n points	trapezoid/Simpson ($n=2,3$)	Newton–Cotes	Gauss
degree	$1,\ 3$	$n-1$ (n even), n (n odd)	$2n-1$

Gauss Weights Are Positive

Proposition. $w_k = \int_{-1}^{1} \ell_k(x)^2\,dx > 0$ for $k = 1, \ldots, n$.

Proof. *1. Exactness applies.* $\ell_k^2 \in \prod_{2n-2} \subset \prod_{2n-1}$, so by the Theorem $G_n(\ell_k^2) = \int_{-1}^{1} \ell_k^2\,dx$.
2. Evaluate the rule. $\ell_k(x_i) = 1$ if $i = k$, 0 otherwise, so $G_n(\ell_k^2) = \sum_i w_i \ell_k(x_i)^2 = w_k$.
3. Sign. $\ell_k^2 \geqslant 0$ is continuous and $\ell_k(x_k) = 1$, so $\int_{-1}^{1} \ell_k^2\,dx > 0$. ■

Remark (*Why it matters: stability*). $f \equiv 1$ gives $\sum_i w_i = 2$, so $\sum_i |w_i| = 2$ for every n. If each $f(x_i)$ carries an error at most δ, G_n changes by at most 2δ. Closed Newton–Cotes rules with 9 points already have negative weights, and for large n the sum $\sum_i |w_i|$ grows without bound.

Error of the Gauss Rule

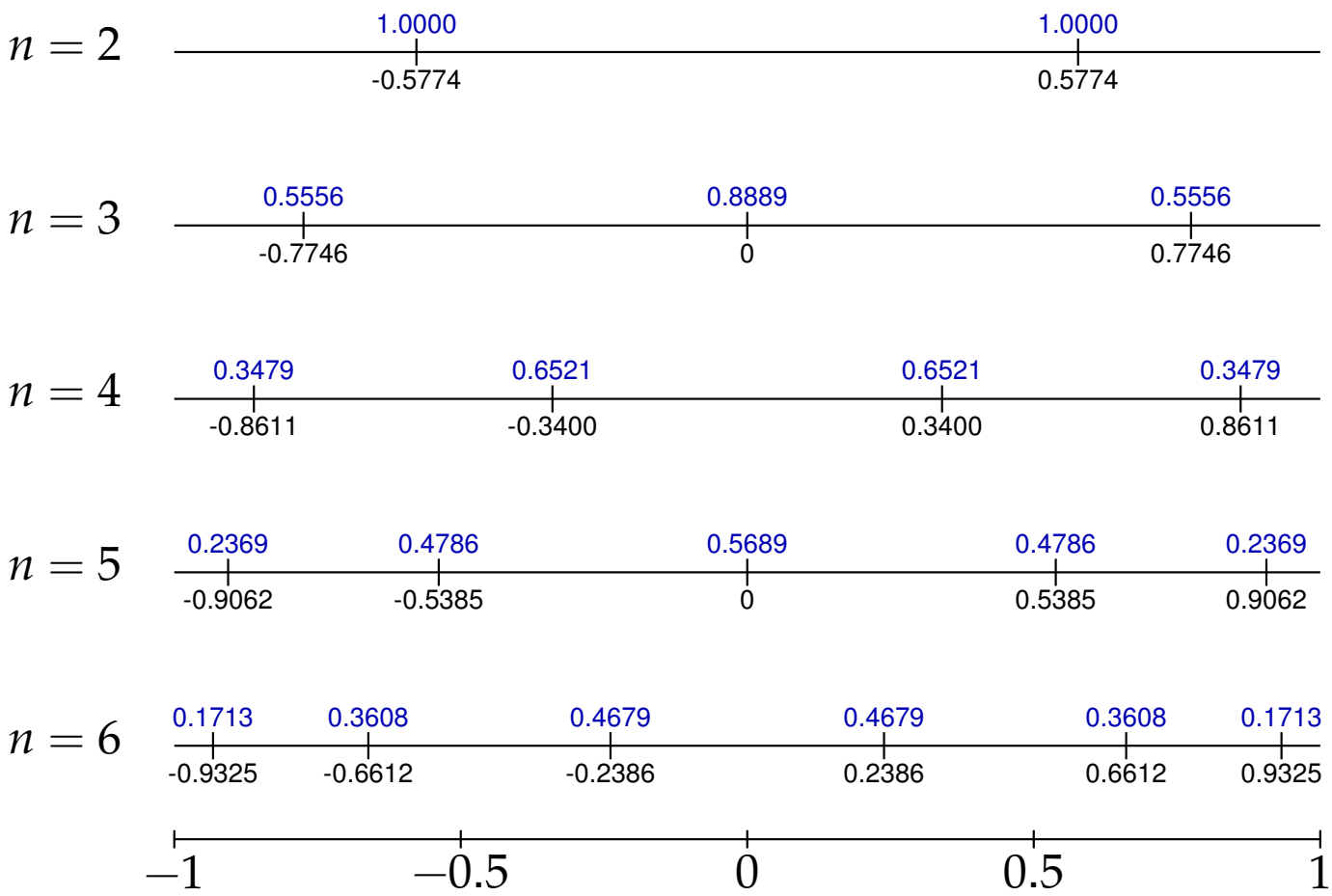
Theorem (*proof omitted; Stoer–Bulirsch, §3.6*). If $f \in C^{2n}[-1,1]$, there is $\xi \in (-1,1)$ with

$$\int_{-1}^{1} f(x)\,dx - G_n(f) = \frac{f^{(2n)}(\xi)}{(2n)!} \int_{-1}^{1} p_n(x)^2\,dx.$$

Example (*Check: $n = 2, f = x^4$*). Left: $\frac{2}{5} - \frac{2}{9} = \frac{8}{45}$. Right: $f^{(4)} \equiv 24$, so $\frac{24}{4!}\int_{-1}^{1}(x^2 - \frac{1}{3})^2 dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$.

- Proof: Hermite interpolation of f, f' at the nodes (degree $2n-1$).
- $f \in C^\infty$: error decreases faster than any power of n.
- On $[a,b]$: apply to $f \circ \varphi$.

Nodes and Weights



Black: nodes. Blue: weights. Symmetric, positive, $\sum_i w_i = 2$. Exact: $n = 3$: $0, \pm\sqrt{3/5}$; $\frac{8}{9}, \frac{5}{9}$.

Matlab

Golub–Welsch: nodes = eigenvalues of a symmetric tridiagonal matrix (why: later, with eigenvalue problems).

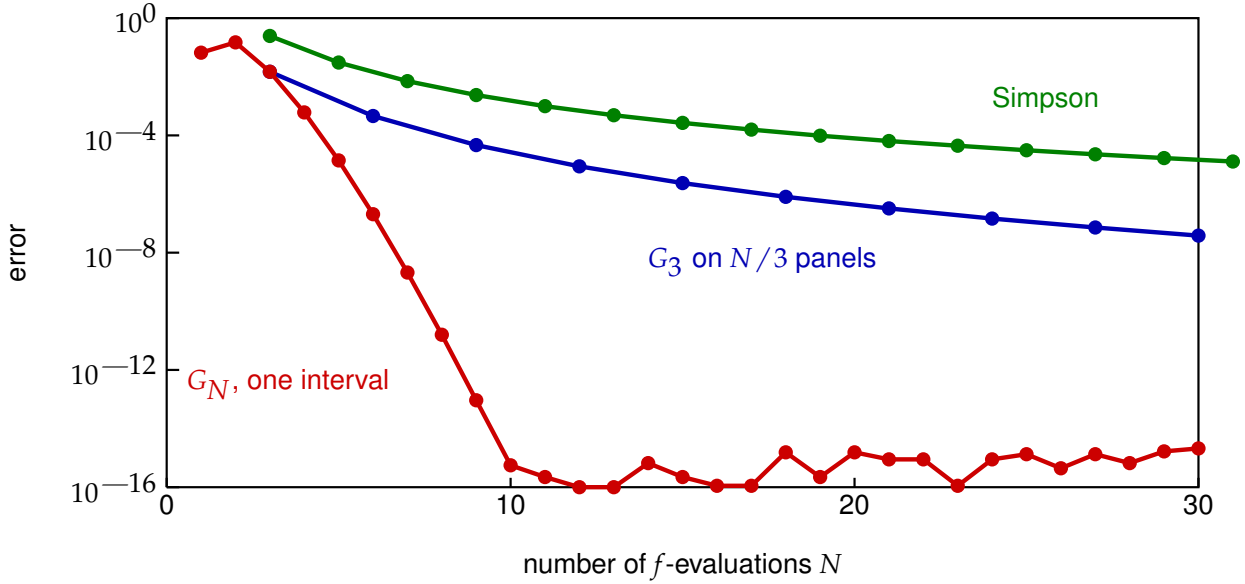
```
function [t, w] = gauss(n)
% n-point Gauss-Legendre nodes t and weights w on [-1,1]
k = 1:n-1;
beta = k ./ sqrt(4*k.^2 - 1);          % recurrence coefficients
T = diag(beta, 1) + diag(beta, -1);    % Jacobi matrix
[V, D] = eig(T);
[t, i] = sort(diag(D));                % nodes
w = 2*V(1, i).^2;                      % weights (row)
```

```
[t, w] = gauss(5);                     % once, outside any loop
x = (a+b)/2 + (b-a)/2*t;               % nodes on [a,b]
Q = (b-a)/2 * w * f(x);                % f vectorized
```

- Compute nodes and weights once (or use a table), never per call.

Experiment

$\int_0^5 xe^{-x}\,dx$: error with N evaluations of f.



- One interval, many nodes > 3 nodes, many panels $>$ Simpson.
- Rounding level (below 10^{-16} drawn at 10^{-16}) by $N = 15$.

Pros and Cons of Gaussian Quadrature

Pros	**Cons**
• Highest degree for n points: $2n-1$. • Positive weights: stable for every n. • No end-point values: works if f is singular at a or b.	• Increasing n moves the nodes (G_2: ± 0.577; G_4: $\pm 0.340, \pm 0.861$): old values cannot be reused. • An error estimate $\lvert G_{2n} - G_n \rvert$ costs $2n$ new evaluations.

Singularity

$\int_0^1 x^{-1/2}dx = 2$. Simpson cannot start ($f(0) = \infty$); G_n runs, all nodes lie in $(0, 1)$.

n	2	4	8	16	32	64
|error|	0.35	0.19	0.10	0.053	0.027	0.014

Error $\approx C/n$: halves when n doubles (f is unbounded, the error theorem does not apply).

Fix: $x = t^2$, $dx = 2t\,dt$ gives $\int_0^1 2\,dt$, exact for G_1.

Beyond One Dimension (Outlook)

Example: $40\,000$ tetrahedra Ω_i, need $\int_{\Omega_i} f\,dx$ on each.

- Map Ω_i to a reference tetrahedron (like φ).
- Gauss-type rule there: $\frac{(n+1)(n+2)(n+3)}{6}$ points.
- Total $\approx 40\,000 \cdot \frac{n^3}{6}$ evaluations: $n = 10$ gives 286 points per tetrahedron, $11.44 \cdot 10^6$ in all.

Summary

rule	f-values	exact on
trapezoid	2	$\prod_1$
Simpson	3	$\prod_3$
Gauss G_n	n	$\prod_{2n-1}$

- Nodes: zeros of P_n; weights $\int \ell_i > 0$.
- Proof: divide by p_n; orthogonality and zeros remove the quotient.
- Smooth f: most accuracy per value. Rough f: rate collapses.

Looking Ahead

- **Next lecture:** Gauss–Kronrod Rules; Adaptive Quadrature